

Design of Online Detection System for Insulation Resistance of Electric Vehicle Based on Unbalanced Bridge

Yongpeng Shen, Ankang Liu, Guangzhao Cui, Gaorui Ge

Department of Electrical and Information Engineering

Zhengzhou University of Light Industry

Zhengzhou 450002, Henan, China

shenyongpeng@zzuli.edu.cn; ankang_zzuli@163.com; cgzh@zzuli.edu.cn; ggr_zzuli@163.com

Abstract—The insulation resistance between the DC bus and the whole vehicle is an important parameter to ensure the safety of the electric vehicle. In view of the low precision, longer measuring time, and poor reliability of the existing insulation detection methods for electric vehicle. The intrinsic mechanism which affects the measurement precision and reliability of insulation resistance is analyzed firstly, and then an insulation resistance detection circuit of unbalanced bridge based which the optical control relay and high-precision instrument amplifier is designed. Furthermore, an improved four-step insulation resistance online calculation approach is proposed. Finally, based on a light electric truck, measurements of multiple sets of insulation resistance values are performance, and the detection error is analyzed under different state of charge and compared with the traditional bridge method. The measurement results show that the measurement error of the proposed electric vehicle insulation detection system has been kept within 5%. What's more, the proposed electric vehicle insulation detection system has advantages of both on-line detection and high reliability.

Index Terms— electrical vehicle, power battery, state of charge, insulation detection, bridge circuit

I. INTRODUCTION

When electric vehicle employs power battery packs as its energy source, the battery voltage is higher than 36V, even can reach hundreds of volts, which far exceeds the range of human safety voltages [1]. The electric vehicle is a complex electromechanical integration product with a series of high-voltage electrical equipment, consisting of the power battery pack, the motor and its controller, the car charger and the auxiliary battery charging device etc. The working condition of these components is comparatively complex, the vibration, acid-base gas corrosion and changes of temperature and

humidity can cause deterioration of the power cables and other insulating materials, and even damage the condition of insulation, which will consequently reduce the insulation strength of the equipment and even endanger the safety of drivers and passengers[2]. Therefore, it must be ensured that the positive and negative terminal of the power battery pack must maintain good insulation status. What's more, one of the basic safety requirements of the electric vehicle is real-time detection of the insulation resistance between the positive and negative terminal of the power battery pack and the vehicle body [3]-[4].

The existing insulation detection technology is typically classified into electric bridge method [5]-[7] and signal injection method [8]. The basic principle of measuring insulation resistance with balanced bridge method is that based on the principle of electric bridge circuit, the insulation resistance of the system is measured by controlling the opening and closing of the relays on the positive and negative terminal of the power battery pack. With the advantages of simple measurement, the method still exists some deficiencies. The most notable defects are low sensitivity and reliability, and mechanical wear caused by mechanical switch relays, as well as lack of consistency and accuracy of the circuit caused by the multiple operational amplifiers.

The basic principle of measuring insulation resistance with the signal injection method is that the voltage signal of a certain frequency is injected between the DC bus and the vehicle body, and the insulation resistance is calculated by measuring the feedback signal. The method can detect both positive and negative terminal insulation resistance at the same time with a high reliability, however, the injection signal will increase the ripple of DC system, and thereby affect the power supply quality and the sampling precision. For example, Ref. [9] proposed an insulation detection method based on capacitor charging and discharging. Its principle is to inject the high voltage into the battery pack, and then the insulation resistance is calculated by the feedback voltage of the capacitor. This method possesses the characteristics of low complexity and high stability, but the maximum absolute measurement error is as high as 12.3%.

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The authors are with the Department of Electric and Information Engineering, Zhengzhou University of Light Industry, Zhengzhou 450002, Henan Province, China. Yongpeng Shen is also with the Henan Key Lab of Information-based Electrical Appliances, Zhengzhou 450002, Henan Province, China

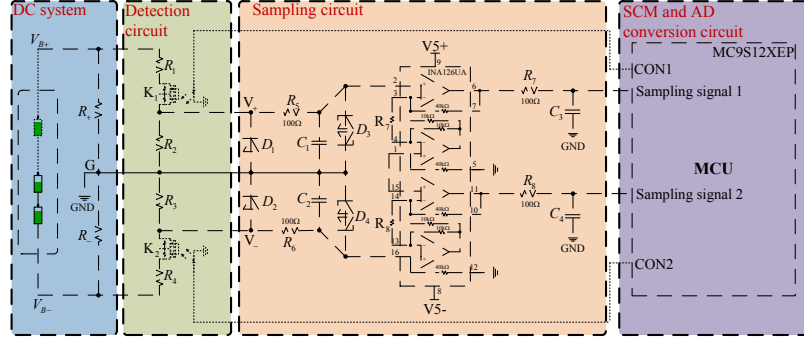


Figure 1. Electric vehicle insulation resistance detection circuit diagram.

In view of the shortcomings of the existing insulation detection method, based on the electric bridge principle, this paper first presents the basic principle of the electric bridge detection method, and then designs the hardware circuit with the employment of optical control relay to control the bridge arm's opening and closing, and the high-precision instrumentation amplifier to enlarge the weak voltage signal and converse voltage polarity at the same time. Based on the basic characteristics of the circuit, the calculation method of four steps insulation resistance of the total pressure measurement, upper bridge arm signal acquisition, lower bridge arm signal acquisition and insulation resistance calculation is designed. Finally, the insulation resistance testing system has been tested, and the detection error is analyzed under different state of charge (SOC), the experimental results verify that the proposed electric vehicle insulation detection system has advantages of both online detection, high precision and high reliability as well as being able to measure the insulation resistance of the positive and negative terminal simultaneously.

II. HARDWARE DESIGN OF INSULATION RESISTANCE DETECTION SYSTEM

A. Overall Function

The insulation resistance detection circuit of the electric vehicle designed in this paper is composed of four parts, including untested DC system, detection circuit, sample circuit, single-chip microcomputer and AD conversion circuit. The circuit schematic diagram is listed in Fig. 1.

B. System Hardware Circuit

The untested DC system consists of the power battery and the possible positive and negative terminal insulation resistances. The positive terminal insulation resistance R_+ is defined as the resistance value between the positive bus of power battery pack and the vehicle ground. Meanwhile, the negative terminal insulation resistance R_- is defined as the resistance value between the negative battery bus of the power battery pack and the vehicle ground. The insulation resistance R_+ and R_- are shown in Fig.1.

The bridge arm sampling resistance R_1 and R_4 are $1880 \text{ k}\Omega$, R_2 and R_3 are $4.7 \text{ k}\Omega$, both of them are the precision resistance with the precision 1%, K_1 and K_2 select the AQV258 optical control MOS relay with voltage withstanding

value of 1500V as its switch, and the model of the clamp diode D_1 and D_2 is CD4148, the two-way TVS diode D_3 and D_4 is PESD5V0S1, the capacity of capacitance C_1 , C_2 , C_3 and C_4 is $0.01 \mu\text{F}$, the resistances of R_5 , R_6 , R_7 and R_8 are all 100Ω .

Selecting dual instrumentation amplifier with the maximum offset voltage $250 \mu\text{V}$, the low offset voltage drift ($3 \mu\text{V}/^\circ\text{C}$ MAX) INA2126 for signal amplification, and achieving voltage polarity conversion simultaneously, it can be completed on the op-amp and collection of the positive and negative signals only with a chip.

Single-chip microcomputer and AD sampling circuit are composed of MC9S12XEP100 MCU and its built-in 12-bit multi-channel AD sampling circuit.

III. PRINCIPLE OF INSULATION RESISTANCE DETECTION FOR BATTERY PACK

A. Analysis of detection methods of traditional bridge method

Detection circuit of insulation resistance of traditional bridge is shown in Fig. 2.

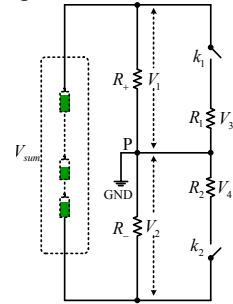


Figure 2. Equivalent circuit of typical bridge circuit.

When using the traditional method, closes the relay K_1 , K_2 , then it can be described as

$$V_1 = \frac{R_+}{R_+ + R_-} V_{\text{sum}} \quad (1)$$

$$V_2 = \frac{R_-}{R_+ + R_-} V_{\text{sum}} \quad (2)$$

Closes K_1 and disconnects K_2 , the formula is as follows:

$$V_3 = \frac{R_+ // R_1}{R_+ // R_1 + R_-} V_{\text{sum}} \quad (3)$$

$$V_4 = \frac{R_-}{R_+ // R_1 + R_-} V_{\text{sum}} \quad (4)$$

Combination with equation (3)(4), it can be obtained as

$$R_+ = \left(\frac{V_4 V_1}{V_3 V_2} - 1 \right) R_1 \quad (5)$$

However, this method has the following problems

- (1) The voltage at both ends of the positive and negative insulation resistance needs to be measured, which leads to the system structure complicated and has a large error;
- (2) There is no solution of the equation when the resistance of the positive or negative insulation resistances is zero.

B. Principle of the improved bridge insulation resistance test method

Based on the insulation resistance detection circuit designed by this paper, its equivalent circuit is shown in Fig. 3. The p point, measuring V_+ and V_- reference 0 potential point, is connected with the vehicle body.

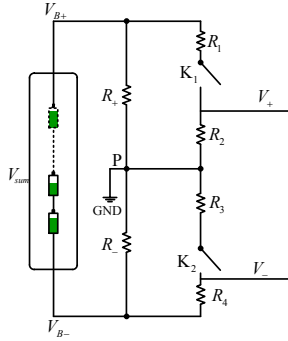


Figure 3. Equivalent circuit of improvement bridge circuit.

The specific detection steps are as follows:

1) measure the total voltage of power battery pack

Taking the negative terminal of power battery pack as reference point, measure the total voltage of the power battery, and close the relay K_1 and K_2 . The equivalent circuit is shown in Fig. 4.

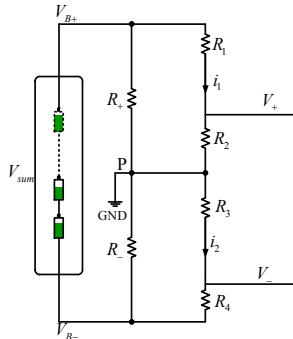


Figure 4. Equivalent circuit when K_1 and K_2 connected.

In fig. 4, i_1 and i_2 are currents flowing through the upper and lower bridges, and the total voltage of the battery pack is the sum of the voltage between the positive terminal and the P point and the voltage between the negative terminal and the P.

$$V_{\text{sum}} = \frac{V_+}{R_2} (R_1 + R_2) + \frac{V_-}{R_3} (R_3 + R_4) \quad (6)$$

2) Measure the sampling point voltage signal in the upper bridge arm

Setting high about SCM control signal CON1, setting low about CON2, now the optical control relay K_1 connect, K_2 disconnect, then selecting P as reference point, measure the upper bridge arm sampling point voltage signal V_+ . The equivalent circuit is shown in Fig.5.

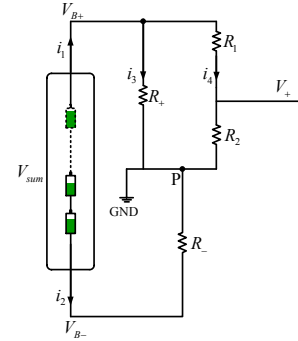


Figure 5. Equivalent circuit when K_1 connected and K_2 disconnected.

The reference current direction of the positive terminal is shown in Fig. 4. The current flowing through R_1 , R_2 can be obtained according to the voltage V_+ detected by measuring the resistance R_2 and p points with the detection circuit

$$i_4 = \frac{V_+}{R_2} \quad (7)$$

Voltage between positive terminal of power battery pack and reference point P.

$$V_{B+} = V_+ \left(1 + \frac{R_1}{R_2} \right) \quad (8)$$

Then it can be concluded that the positive current of the power battery pack. (current i_3 is the current flowing through the positive terminal insulation resistance R_+), it is derived as

$$i_1 = i_3 + i_4 = \frac{V_{B+}}{R_+} + \frac{V_+}{R_2} = V_+ \left(\frac{1}{R_+} + \frac{R_1}{R_+ R_2} + \frac{1}{R_2} \right) \quad (9)$$

The direction of the negative reference current is shown in the diagram, $V_{B+} + V_{B-} = V_{\text{sum}}$, so the basic circuit principle is $i_1 + i_2 = 0$.

$$(R_1 + R_2) R_- + R_+ R_- = R_2 \frac{V_{B-}}{V_+} R_+ \quad (10)$$

3) Measure the sampling point voltage signal in the lower bridge arm

Setting high about control signal CON1, low about CON2, right now the optical control relay K_2 connect, K_1 disconnect, then selecting P as reference point, measure the upper bridge arm monitoring point voltage signal V_+ . The equivalent circuit is shown in Fig. 5.

As the circuit changes, the actual voltage of the reference point P changes, so the voltage and current of the positive and negative poles change with respect to the reference zero potential point, which is similar to step (2).

$$(R_3 + R_4)R_+ + R_+R_- = R_3 \frac{V_{B+}}{V_-} R_- \quad (11)$$

4) Insulation resistance calculation

Through the above steps, the sampling point voltages V_+ and V_- are measured and divided into the following four cases according to different voltage.

case 1:

When V_+ and V_- are equal to zero, both the positive insulation resistance R_+ and the negative insulation resistance R_- are equal to infinity.

case 2:

When V_+ is equal to zero and V_- is not equal to zero, the negative insulation resistance R_- is equal to infinity and the relays K_1 disconnect and K_2 connect. The equivalent circuit is shown in Fig. 6.

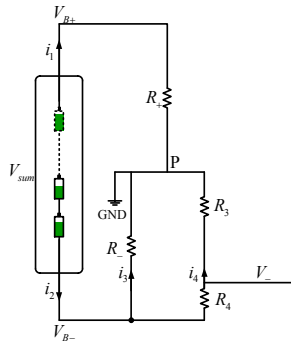


Fig. 6. Equivalent circuit when K_2 connected and K_1 disconnected.

Flowing through the resistance R_+ current $i_1 = i_4$, which can be described as

$$R_+ = \frac{V_{B+}}{i_1} = \frac{V_{sum} - \frac{V_-}{R_3} (R_3 + R_4)}{\frac{V_-}{R_3}} \quad (12)$$

case 3:

When V_+ is not equal to zero and V_- is equal to zero, the negative insulation resistance R_- is equal to infinity and the relays K_1 connect and K_2 disconnect. The equivalent circuit is shown in Fig. 5.

Flow through the negative terminal insulation resistance R_- current $i_2 = i_4$, thus

$$R_- = \frac{V_{B-}}{i_2} = \frac{V_{sum} - \frac{V_+}{R_2} (R_1 + R_2)}{\frac{V_+}{R_2}} \quad (13)$$

case 4:

When V_+ and V_- are equal to zero, the insulation resistance calculation process is as follows:

It can be configured by Eq. (10) and Eq. (11)

$$R_- = AR_+ \quad (14)$$

Where $A = (R_1 + R_2 + R_3 + R_4) / (R_2 \frac{V_{B-}}{V_+} + R_3 \frac{V_{B+}}{V_-})$

$$R_+ = \frac{R_1 + R_2 - R_2 \frac{V_{B-}}{V_+} A}{A} \quad (15)$$

IV. SOFTWARE DESIGN IF INSULATION RESISTANCE DETECTION SYSTEM

The software design was performed in Freescale's Code warrior programming conditions, by which the functions of control of light-controlled relay, voltage signal sampling, total voltage measurement of power battery, classification and calculation of insulation resistor resistance, fault alarm and fault handling were realized.

According to the national standard GB/T18384.3-2015, the insulation resistance should be greater than $100 \Omega/V$ [1]. Therefore, $500 \Omega/V$ are selected as the safety threshold. Due to the introduction of the detection resistance, the insulation between the power battery and the vehicle body reaches a lower level, which affects the accuracy of detection. In order to ensure that the detection resistance does not affect the performance of the insulation detection system, the relays K_1 and K_2 are closed only at the instant of detection. After the end of the detection, the relays are in the state of disconnection, and the measurement bridge arm is disconnected with the positive and negative ends of the power battery.

Insulation resistance calculation process diagram is shown in Fig. 7.

V. EXPERIMENTAL RESULTS AND ANALYSIS

In order to test the performance of the power battery insulation resistance detection system, an experimental platform was set up in the laboratory as shown in Fig. 8.

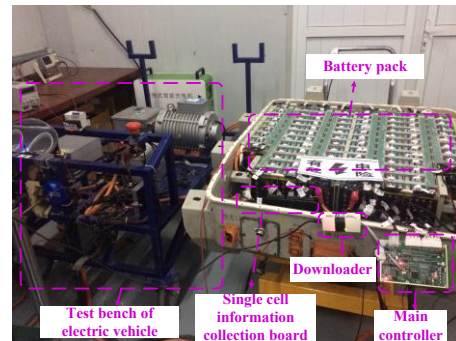


Figure 8. Insulation detection of experimental platform.

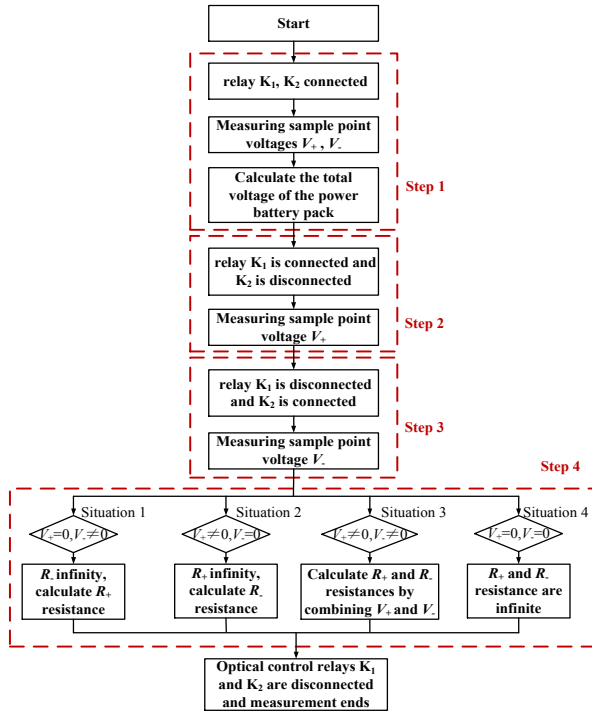


Figure 7. Insulation resistance calculation process diagram

The power battery used in the experiment is an 18650 ternary lithium battery with a nominal value of 3.6V and a capacity of 2.6Ah. The 45 batteries in parallel as a section, a total of 90 sections in series composed of power battery pack, and the nominal voltage and capacity of power battery pack are 330V, 117Ah.

Different resistance values are simulated by connecting different resistance resistors in parallel between the two ends of the power battery and the vehicle body. The measurement results are sent to the upper computer display interface through the CAN bus.

Compared the experimental results with the results measured by the FLUKE 1587C model insulation resistance meter, the test results for the three groups with different insulation resistances are shown in Tab. I.

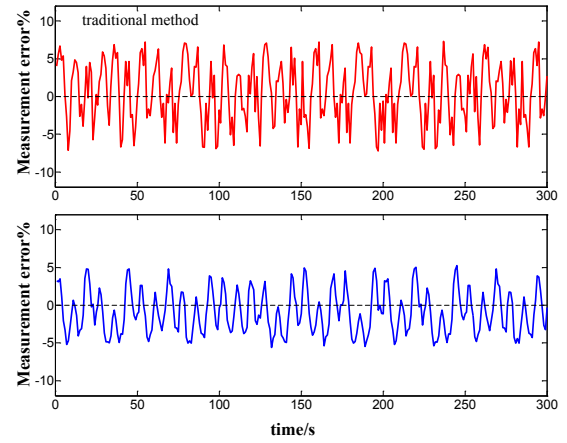
TABLE I. TEST RESULTS

Test items	Test Results		
	1	2	3
R_+ Actual value	1 M Ω	2 M Ω	1 M Ω
R_- Actual value	1 M Ω	1 M Ω	1 M Ω
Detection voltage V_+	0.335V	0.3906V	0.5311V
Detection voltage V_-	0.3152V	0.1968V	0V
R_+ Measurement	0.971 M Ω	1.941 M Ω	∞
R_- Measurement	0.965 M Ω	0.962 M Ω	0.973 M Ω
FLUKE measurement R_+	0.988 M Ω	1.975 M Ω	∞
FLUKE measurement R_-	0.988 M Ω	0.987 M Ω	0.987 M Ω

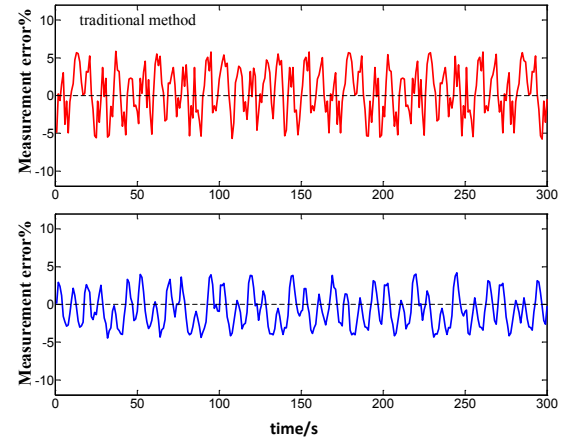
The experimental results show that the designed electric vehicle insulation resistance detection circuit and control

method can detect various insulation resistance conditions occurring at the positive and negative terminals. According to the data in Tab. 1, the measurement error of the insulation resistance of the detection system is kept within 5 %, which can suffice the insulation testing requirements of pure electric vehicles.

In the test, a 2M Ω resistance is connected between the battery pack and the car chassis. The test results of insulation resistance for a period of time are obtained by CAN bus, and the absolute error waveform as shown in Fig. 9.



(a) Error curves in 10% SOC



(b) Error curves in 10% SOC

Figure 9. test error waveform with different SOC.

It can be observed from Fig.9 that the measurement error is different with different residual electricity, which is caused by the change of the battery resistance and the battery voltage. The lower of the battery residual electricity quantity, the larger of the internal resistance, the total battery pressure will be reduced. Therefore, the measurement error of insulation resistance with a low residual power is slightly higher than that of insulation resistance with enough battery residual power.

VI. CONCLUSION

To make up for the shortcomings of the traditional insulation detection method, an electric vehicle insulation resistance detection circuit based on the optical control relay

and a control method are designed, in addition, the proposed four-step insulation resistance calculation method can be applied to a variety of fault situations. Experiments under multiple sets of insulation resistance values are carried out, the results show that the electric vehicle insulation resistance detection system developed in the vehicle operation can on-line measure the different insulation resistances on the positive and negative sides, and the test results are stable and the measurement error has been kept within 5% during the vehicle running process. What's more, the measurement accuracy of insulation resistance under different SOC is measured, and the experimental results show that the amount of residual power has certain influence on the measuring precision of insulation resistance.

REFERENCES

- [1] China Quality Press, GB/T 18384.1-2015. Electrically propelled road vehicles safety specifications Part 3: Protection electric shock protection.
- [2] M. McDonough, "Integration of Inductively Coupled Power Transfer and Hybrid Energy Storage System: A Multiport Power Electronics Interface for Battery-Powered Electric Vehicles," *IEEE Transactions on Power Electronics*, vol. 30, no. 11, pp. 6423-6433, 2015.
- [3] R. Rybski, J. Kaczmarek and K. Kontorski, "Impedance Comparison Using Unbalanced Bridge With Digital Sine Wave Voltage Sources," *IEEE Transactions on Instrumentation and Measurement*, vol. 64, no. 12, pp. 3380-3386, Dec. 2015.
- [4] H. Torkaman, F. Karimi, "Influence of ambient and test conditions on insulation resistance/polarization index in hv electrical machines - a survey," in *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 22, no. 1, pp. 241-250, 2015.
- [5] L. Bi, X. Wei, Z. Sun, "A High-voltage Safety Protection Method for Electric Vehicle Based on FPGA," *IEEE International Conference on Vehicular Electronics and Safety, ICVES*, pp. 26-31, 2006.
- [6] T. Goh, M. Park, G. Koo, M. Seo and S. W. Kim, "State-of-health estimation algorithm of Li-ion battery using impedance at low sampling rate," *2016 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC)*, Xi'an, pp. 146-150, 2016.
- [7] X.X. Song, et al., "Insulation resistance monitoring algorithm for battery pack in electric vehicle based on extended Kalman filtering," *Energies*, vol.10, no.5, 2017.
- [8] X. Zhou and S. Chen, "Study on insulation detection method of electric vehicles based on single point of failure model," *11th International Forum on Strategic Technology, IFOST 2016*, pp. 191-194, 2017.
- [9] Mathsyaraja, Avarind, "Ground fault detection for flexible high voltage power systems," *SAE Int. J. Commer. Veh.* vol. 4, pp. 185-191, 2011.